



OpenGL-ES 1.1 Ext. Compliant 3D Graphic Accelerator for Mobile Phone, PDA

Features

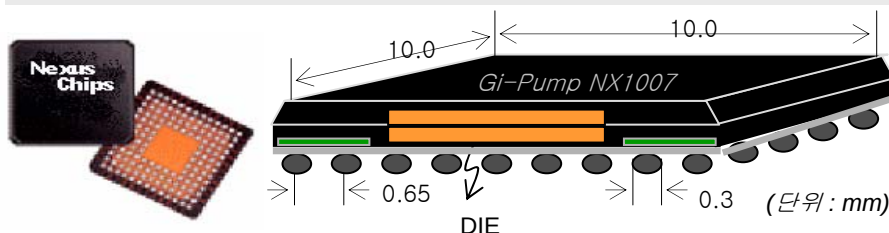
1. 3D Graphic Functions

- Fully Compliant with OpenGL ES 1.1 Ext. Common
- Fully Compliant with JSR-184, Mobile JAVA 3D
- Direct3D Mobile support
- 12.5 million triangles/sec@100MHz
- 16/18/32-bit asynchronous I/F for base band processors
- 32-bit SMI (System memory Interface)
- High speed h/w key frame animation (Morphing)
- H/W Bone animation / Point sprite
- Supports direct mode and buffer objects
- Full Hardwired Geometric / Rendering engine
- Supports lighting up to 8 light sources
- Flat/Gouraud Shading
- Supports point/spot/directed light sources
- Projection & Model view transform engine
- Perspective division
- View port transform
- Culling
- Hardware clipping checking and high speed internal clipping processing
- Supports User Clip plane
- Texture mapping
Min. Filtering : mipmap, pseudo-mipmap
Mag. Filtering : Bilinear
Wrapping (Clamping, Repeat, Mirror)
- Frame buffer object
- Fill mode

- Supports Compressed Texture(palette, DXT1, DXT3)
- Supports Multi-Texture
- Texture data format and copy
8 bits Alpha/8 bits Luminance/Luminance-Alpha 88 /RGB 565/RGBA 5551/RGBA 4444
- Texture Combining
Replace / modulate / decal / blend / add / combine
- Fog processing
- Blending (supports also blend with fixed alpha value and extended blending)
- Depth test with 8 modes (up to 16 bits)
- Stencil test with 8 modes (up to 4 bits)
Stencil extensions (keep, zero, replace, increment, decrement, incr_wrap, decr_wrap)
- Alpha test with 8 modes
- Scissor test
- Transparent color
- Mask function (Stencil, depth, color mask)
- Logic operations with 16 modes
- Supports Landscape contents

2. 2D Graphic Functions

- Supports SVG, GIGA2D
- Supports Primitives
- BitBLT with Transparent Color, Scaling and Rotate
- Supports ROP3 operations
- Supports Alpha blending operations
- Supports Sprite engine
- Supports Landscape concept
- Effective block data transferring among memories



Package Type	QVGA/CSP
Pin Number	169 balls
Body Size	10.0mm x 10.0mm
Height	1.4
Memory	Stacked/Extensible

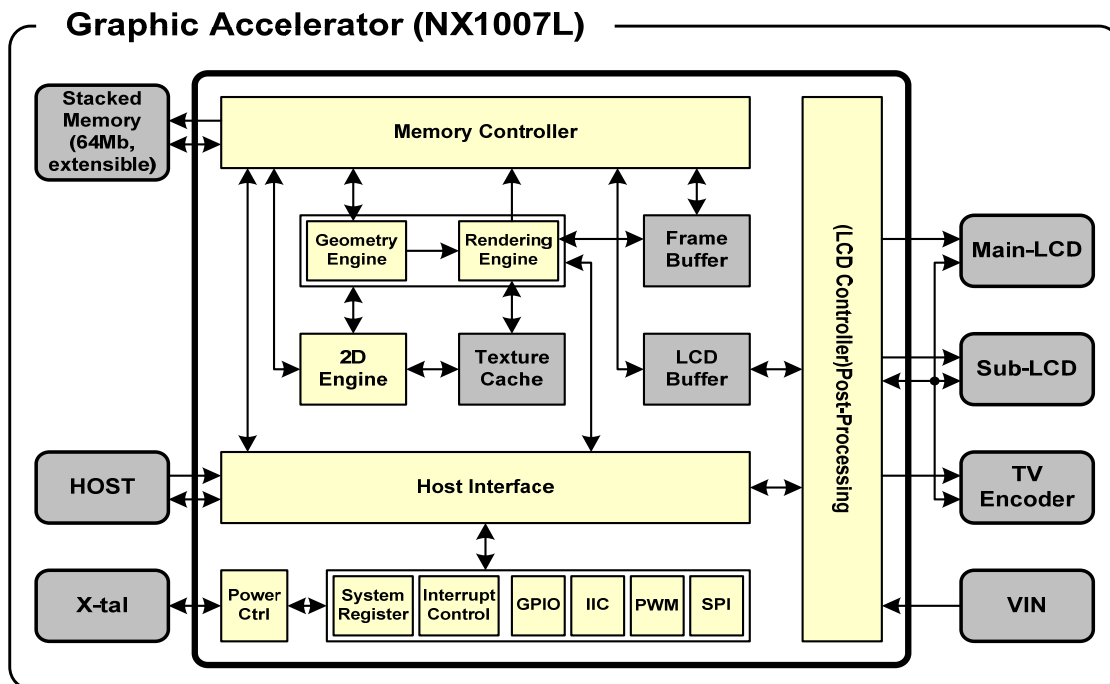
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Features are subject to change without notice.

Overview



3. Display Control

- Supports Double Buffering
- Up to 600 x 800 Display (SVGA)
- Display Rotation
- Anti-aliasing Filtering
- CPU-type LCD Interface
- Support NTSC/PAL encoder

4. Performance

- 12.5M Polygons/second @ 100MHz
- Pixel Fill rate 200M pixels/second @ 100MHz
- User programmable run-time clock change
- Highly integrated low cost solution
- Low power consumption (100mA @ 65MHz) - TBD
- Power save mode supports: all/sub-block power down (under 3mA) - TBD
- Power down mode supports: Stacked Memory & PLL Power down & all power down mode (under 150uA) - TBD
- On chip PLL (Input range = 10MHz~40MHz, output range = 50MHz ~ 250MHz)
- On chip Regulator for internal PLL and core voltage

5. Application, etc

- 169 balls CSP uBGA package (10mm x 10mm)
- Supports Device Driver Software
- Supports Rich Graphic Libraries
- Directly Applicable to Cellular Phone, Smart Phone, PDA or Hand held Device